

Worksheet 7D: Markov Chains — Probability Meets Matrices

Series 7: Linear Algebra
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What This Worksheet Is About

A Markov chain is a system that jumps between states randomly, where the probability of the next state depends only on the current state. When you write the transition probabilities as a matrix and multiply by a probability vector, matrix multiplication is doing real predictive work. Markov chains power Google's PageRank, autocomplete, financial models, and the inference inside language models.

Part A: States, Transitions, and Transition Matrices (10 problems)

A **transition matrix** P has entry P_{ij} = probability of moving from state i to state j . Every row must sum to 1. A **state vector** $\pi^{(t)}$ is a row vector of probabilities over states at time t . The update rule is:

$$\pi^{(t+1)} = \pi^{(t)} P$$

Dublin Weather Example: $P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$ (rows: today Sunny / Rainy; columns: tomorrow Sunny / Rainy)

- Using the Dublin weather matrix: (a) Probability of rain tomorrow if today is rainy? (b) Probability of sunshine tomorrow if today is rainy? (c) Verify each row sums to 1.
- A website has pages Home (H), Products (P), Contact (C). Transition matrix: $P = \begin{bmatrix} 0.2 & 0.5 & 0.3 \\ 0.1 & 0.6 & 0.3 \\ 0.4 & 0.4 & 0.2 \end{bmatrix}$. (a) Probability of visiting Products from Home? (b) Probability of staying on Contact? (c) Verify rows sum to 1.
- A student studies (S) or procrastinates (P). $P(\text{continues studying})=0.8$; $P(\text{starts studying from procrastinating})=0.6$. Write the 2×2 transition matrix.
- Which can be a row of a transition matrix? (a) $[0.5 \ 0.5]$ (b) $[0.3 \ 0.8]$ (c) $[1.0 \ 0.0]$ (d) $[0.3 \ 0.4 \ 0.3]$ (e) $[0.2 \ -0.1 \ 0.9]$
- Start with $\pi^{(0)} = [0 \ 1]$ (definitely rainy) and the Dublin weather matrix. Compute $\pi^{(1)}$ and $\pi^{(2)}$.
- Start with $\pi^{(0)} = [0.5 \ 0.5]$. Compute $\pi^{(1)}$ and $\pi^{(2)}$.

7. Interpret $\pi^{(2)}$ from problem 5 in plain English.
8. Why do we write $\pi^{(t+1)} = \pi^{(t)} P$ (row vector on left) rather than a column-vector convention? What would change with the column convention?
9. Using the study/procrastination matrix from problem 3, start at $\pi^{(0)} = [1 \ 0]$ (studying). Compute $\pi^{(1)}$ and $\pi^{(2)}$.
10. A fair coin has states H and T, each transition probability 0.5. (a) Write the transition matrix. (b) Why does the Markov property hold trivially for a fair coin?

Part B: Powers of the Transition Matrix (8 problems)

The n -step transition probabilities are given by P^n . Starting from $\pi^{(0)}$:

$$\pi^{(n)} = \pi^{(0)} P^n$$

11. Compute P^2 for the Dublin weather matrix.
12. Using P^2 : if today is definitely sunny, what is the probability of sunshine in two days?
13. Verify by computing $\pi^{(0)} P^2$ with $\pi^{(0)} = [1 \ 0]$.
14. Also check via the two-step method $\pi^{(0)} \rightarrow \pi^{(1)} \rightarrow \pi^{(2)}$.
15. Complete the table for $\pi^{(0)} = [1 \ 0]$:

n	Sunny	Rainy
0	1.000	0.000
1	0.700	0.300
2	_____	_____
3	_____	_____
4	_____	_____

16. What do you notice as n increases? Is the distribution converging?
17. Repeat the table starting from $\pi^{(0)} = [0 \ 1]$ (definitely rainy). Compare rows 3–4 with problem 15.
18. This convergence regardless of starting state is a key property. What does it suggest about long-run predictions?

Part C: The Steady State (10 problems)

A **stationary distribution** π satisfies $\pi P = \pi$ with entries summing to 1. For a 2×2 chain with $P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}$:

$$\pi = \left[\frac{b}{a+b}, \frac{a}{a+b} \right]$$

19. For the Dublin chain ($a = 0.3$, $b = 0.4$), find π .
20. Verify $\pi P = \pi$ directly.

21. Does the long-run distribution in your table from problem 15 approach π ?
22. For the study/procrastination chain (problem 3), solve $\pi P = \pi$, $\pi_S + \pi_P = 1$ for the stationary distribution.
23. Interpret the answer to problem 22 in plain English.
24. For the website chain (problem 2), explain in plain English why the stationary distribution is useful for a website designer.
25. For $P = \begin{bmatrix} 1 & 0 \\ 0.3 & 0.7 \end{bmatrix}$, state 1 is absorbing. (a) Will the chain always end in state 1? (b) What is the stationary distribution?
26. A random walk on Left (L), Centre (C), Right (R): from C, go L or R with probability 0.5 each; L and R are absorbing. Write the transition matrix and describe the long-run behaviour.
27. **PageRank:** A random web surfer's stationary distribution gives each page's "importance." Why does a page rank higher if many other high-ranked pages link to it?
28. If every row of P is identical ($P_{ij} = \pi_j$ for all i), what is the stationary distribution? Is this interesting?

Part D: Applications (8 problems)

29. **Simplified PageRank.** Three pages: A links to B and C equally; B links only to A; C links to A and B equally. (a) Write the 3×3 transition matrix. (b) Verify rows sum to 1. (c) Which page do you expect to rank highest, and why?
30. **Language model.** Transition matrix for {I, like, cats, dogs}: $P = \begin{bmatrix} 0 & 0.5 & 0.2 & 0.3 \\ 0 & 0 & 0.6 & 0.4 \\ 0.8 & 0.1 & 0 & 0.1 \\ 0.7 & 0.1 & 0.1 & 0.1 \end{bmatrix}$.
(a) Most likely word after "I"? (b) Probability of "I like cats"? (c) Where does P come from in a real model?
31. **DNA.** A doubly stochastic transition matrix (each row and column sums to 0.25) over bases {A,T,G,C}. What is the stationary distribution? What does it tell you?
32. **Labour market.** $P = \begin{bmatrix} 0.92 & 0.05 & 0.03 \\ 0.30 & 0.50 & 0.20 \\ 0.10 & 0.15 & 0.75 \end{bmatrix}$ (Employed, Unemployed, Inactive). (a) Probability an employed person loses their job? (b) Probability an unemployed person finds work? (c) Interpret $P_{32} = 0.15$.
33. **Mixing time.** What features of P might slow convergence to the steady state?
34. **RL connection.** Why is the Markov property essential for reinforcement learning algorithms to work?

35. Is $P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.3 & 0.7 \\ 0.6 & 0 & 0.4 \end{bmatrix}$ ergodic? Trace paths between all state pairs.
36. Reflection: (a) Where does the row-vector times matrix operation from ws07a appear here? (b) How does a Markov chain extend basic probability (ws06a) to sequences? (c) Why does $\boldsymbol{\pi}P = \boldsymbol{\pi}$ resemble an eigenvector equation?